Akarsh Kudrimoti

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Education

Georgia Institute of Technology — Atlanta, GA

B.S. Computer Engineering (Distributed Design and Systems Architecture) — *Expected May 2028*
Minor in AI/Robotics

- GPA: 4.0/4.0 | ACT: 35
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Deep Learning, Machine Learning

Experience

NEXS Research Group, Georgia Tech — Aug 2026 – Present

Undergraduate Researcher – Agentic LLM Serving Systems — *Atlanta, GA*

- Researching serving systems for agentic LLM workloads on Aragog in Python: scheduling and batching graphs of tool calls, sub-agents, and retries so multi-step agents answer at interactive latency
- Evaluating continuous batching and KV-cache reuse across an agent's repeated model calls in vLLM-class inference engines to cut redundant prefill compute, applying lessons from building Crawfish
- Profiling multi-agent traces to find where serving time goes (prefill, decode, tool-call waits) so scheduling work targets the stages that actually dominate end-to-end latency
- Studying failure handling mid-plan (tool-call retries, parallel branches over shared state) so agent graphs finish instead of restarting from scratch when one call fails

Alphius — May 2025 – Present

Founding Engineer (Financial risk analysis startup) — *Remote*

- Built Product Diligence.AI, a Python financial-risk scoring engine that earned a $2M-valuation term sheet
- Developed NLP data pipelines with spaCy, Spark, and PostgreSQL ingesting 10,000+ pages/day of filings
- Added caching to Azure-hosted Django GraphQL services, cutting resolver latency 40% under production load

Stride Labs — Jan 2025 – May 2025

Software Engineering Intern (Scan-to-insole orthotics startup) — *Atlanta, GA*

- Cut order turnaround 25% by building a responsive HTML/CSS + React UI and Node.js ordering system on PostgreSQL
- Built a Three.js viewer rendering 3D foot scans into orthotic models, cutting clinician fit-review time 90%
- Secured patient PHI across 5 REST API routes with a zero-trust auth layer on Firebase Auth and GCP IAM

Projects

Crawfish: Agent Orchestration Platform | *TypeScript, Node.js, Express, Prisma, MCP* — Sep 2025 – Present

- Shipped an open-source (GitHub) native macOS app wiring 300 services behind a Tauri 2 shell over a localhost REST and SSE server
- Built a semantic codebase-navigation optimizer that cut agent token use 3.25x versus blind grep and file reads
- Designed 50 MCP servers exposing board and filesystem tools so agents pick up tasks and read or write files

LLM Infrastructure Home Lab | *Docker, Kubernetes, vLLM, CI/CD, Prometheus, Grafana* — Aug 2025 – Present

- Built a self-hosted GPU server serving local LLM inference at 45 tok/s via vLLM and Ollama for 5 users
- Configured a Docker and Kubernetes (k3s) cluster orchestrating 3 models, monitored via Prometheus and Grafana
- Built automated LoRA fine-tuning pipelines that cut training cost roughly 60% versus renting cloud GPUs

Redis Clone with Vector Search | *C++17, epoll, KNN, Quantization* — Jan 2026 – Present

- Built an in-memory vector store in C++ from scratch with an epoll event loop and a custom open-addressing map
- Added KNN cosine-similarity search, then cut memory per vector 4x with int8 quantization at minimal recall loss

Mesh Search: Distributed Semantic Search | *Rust, TCP, Noise Encryption* — 2026 – Present

- Built a distributed search engine in Rust across a multi-hop node mesh with BFS query flooding, reciprocal rank fusion of ranked results, and end-to-end encrypted links via the Noise protocol
- Implemented leader election from gossiped capability scores with idempotent lease claims and 5–7s failover
- Instrumented every link event and query as JSONL telemetry and built a terminal dashboard reading per-query latency, bytes per link, and mesh reach off the live stream

Skills

Web: React, Node.js, TypeScript, JavaScript, Express, GraphQL, REST APIs, Three.js, Tauri, Prisma, HTML/CSS
Infrastructure: PostgreSQL, SQL, Docker, Kubernetes (k3s), Azure, GCP, Firebase, Git/GitHub, CI/CD, Prometheus, Grafana
Languages & AI: TypeScript, JavaScript, Python, C/C++ (C++17), Rust, SQL; LLM/GenAI integration, MCP, agentic systems, AI-assisted development
Awards: Zell Miller Scholar, USACO Gold, Georgia Tech InVenture Prize Finalist